

Disclaimer

- This training resource was developed to be used with the CZ ID platform
- In October, 2026, stewardship of CZ ID transitioned to UCSF and the platform was renamed to SeqtoID
- These training materials are not being actively maintained and may not reflect differences between CZ ID and SeqtoID, subsequent changes to the platform, or current methods and best practices in metagenomic sequencing and analysis



CZ ID mNGS Nanopore Pipeline

mNGS Nanopore Pipeline Highlights

- Developed in collaboration with bioinformatics team from Oxford Nanopore Technologies (ONT)
- Pipeline accurately identifies microbes from a range of samples:
 - Mock microbial communities
 - Cell culture
 - Humans (stool, nasal samples)
 - Insects (mosquitos)
 - Environmental (hot springs)

mNGS Workflow Overview

Sample
Collection

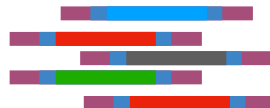
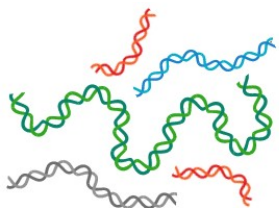
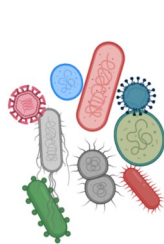
DNA or RNA
Extraction

Library
Preparation

Sequencing

Bioinformatic
Analysis

mNGS



- **Taxonomic Identification**
- Microbiome analysis
- Discovery

Focus for this workshop

Pathogen identification in stool samples and nasopharyngeal swabs

Sequence File Formats: FASTA vs FASTQ

>Nucleotide Sequence Name

```
ATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGG  
CTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATT  
GGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCTAATTGGCT
```

```
@NB501961:14:HM7TLBGX2:1:11102:3233:17234 1:N:0:GATCACCA+GATCACCA  
CATTCGGCTGGGTTTCGTCACCCTGCGGGAAGATGCGGGTCCAGGCGATAGAGGTGCGGAAGCATTGAAGCCCA  
+  
AAAAAEEEEEEEEEE/EEEEEEEEEEEEEEEEEEEE<EEEEEEEE//EAEEEEEEAA/EEE/EEAAEEE<6/
```

Sequence File Formats

- General
 - FASTA
 - FASTQ (Includes sequence identifier, sequence, and **quality scores** for base calls)
- Nanopore-specific formats
 - FAST5
 - POD5

Nanopore Specific formats

Feature	POD5	Older FAST5
File format	Binary (Apache Arrow-style, columnar)	HDF5-based (hierarchical)
Speed	High (parallel I/O optimized)	Slower with random access issues
Size	Smaller (less overhead)	Larger
Basecaller support	Dorado, Guppy (via pod5-api)	Guppy, older tools
Ecosystem	Actively maintained by ONT	Being phased out

mNGS Nanopore Pipeline Workflow

Upload & Data Pre-processing

Nanopore data
(FASTQ files)

Notes:

- Pipeline only accepts **FASTQ** files (FAST5 files must be converted)
- You can upload files through the web app or [command line interface \(CLI\)](#)
- The ONT pipeline does not complete adapter/barcode trimming, so FASTQ files must be trimmed prior to upload
- Only one file per sample:
Multiple FASTQ files for a given sample will be automatically concatenated if filenames follow a certain format and are associated with the same barcode (see [Automatic Concatenation of Nanopore Files](#) for details)

If filenames do not follow the format recognized by the platform, you need to [concatenate](#) (or combine) them into a single file before upload
- Reads with midread barcodes detected should be filtered prior to upload (see more on the Prepare ONT fastqs for upload to CZ ID slide deck)

mNGS Nanopore Pipeline Workflow

Upload & Data Pre-processing

Nanopore data
(FASTQ files)

Quality, Host &
Human Filtering
(fastp &
minimap2)

Subsampling
to 1M reads



- Adapter trimming occurred on the basecaller software
- Minimum Q ≥ 9
- Length ≥ 100
- Complexity > 30%
- Host filtering is splice aware if input is RNA

mNGS Nanopore Pipeline Workflow

Upload & Data Pre-processing

Assembly & Mapping

Nanopore data
(FASTQ files)

Quality, Host &
Human Filtering
(fastp &
minimap2)

Subsampling
to 1M reads

Assembly
(metaFlye)

Contigs

Contigs



Notes:

- metaFlye only assembles reads > 1kb
(in many cases we don't see contigs)

mNGS Nanopore Pipeline Workflow

Upload & Data Pre-processing

Assembly & Mapping

Nanopore data
(FASTQ files)

Quality, Host &
Human Filtering
(fastp &
minimap2)

Subsampling
to 1M reads

Assembly
(metaFlye)

Contigs

Mapping reads to
contigs
(minimap2)

Subtraction of
contig-associated
reads
(Samtools)

Unassembled
reads

Contigs



Un-assembled reads



mNGS Nanopore Pipeline Workflow

Upload & Data Pre-processing

Assembly & Mapping

Species Identification

Nanopore data
(FASTQ files)

Quality + Host
Filtering
(fastp &
minimap2)

Subsampling
to 1M reads

Assembly
(metaFlye)

Subtraction of
contig-associated
reads
(minimap2)

Contigs

Unassembled
reads

Contig alignment against **NT
and NR** databases
(minimap2 & DIAMOND)

Unassembled read alignment
against **NT** database
(minimap2)

Contigs
aligned to NCBI NT and NR

Aggregate

Un-assembled reads
aligned to NCB NT only

Species A

Species B

Species C

mNGS Nanopore Pipeline Workflow

Upload & Data Pre-processing

Nanopore data
(FASTQ files)

Host + Quality
Filtering
(minimap2)

Subsampling
to 1M reads

Assembly & Mapping

Assembly
(metaFlye)

Contigs

Subtraction of
contig-associated
reads
(minimap2)

Unassembled
reads

Species Identification

Contig alignment against **NT**
and NR databases
(minimap2 & DIAMOND)

Unassembled read alignment
against **NT** database
(minimap2)

Merge
results

(Aggregate
total bases
per taxon)

Report

Sample Report

Things to note

- mNGS Nanopore pipeline is limited compared to the Illumina pipeline
 - No QC visualization on CZ ID
 - No background models on CZ ID (more on that later)
 - No consensus genomes on CZ ID
 - No visualization to compare samples (heatmap, trees)

Heatmap and background model alternative: [No-code interactive app accessed via github](#)

CZ ID mNGS Pipeline for ONT review complete!

- Next: QC and Interpretation for ONT

Resources

- Preprint describing the mNGS Nanopore pipeline and its implementation.
 - [CZ ID: a cloud-based, no-code platform enabling advanced long read metagenomic analysis](#)
- See “how-to” guides in CZ ID’s Help Center:
 - [Upload Nanopore Data](#)
 - [Analyze Nanopore Data](#)
- To learn more about the pipeline, see [Nanopore Pipeline Validation](#) or click [here](#) to view pipeline changes over time.
- “Mid-read” [barcode trimming tool](#)
- [Heatmap visualization and background subtraction tool](#)